

CUSTOMER INTELLIGENCE REPORT

SQL-Driven Customer Analytics & Loyalty Intelligence Dashboard

Submitted By

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Tools & Technologies Used

- Python
 - Pandas
 - NumPy
 - Matplotlib
 - Seaborn
 - MySQL
 - Power BI
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1. Project Overview

The objective of this project was to analyze customer purchasing behavior and generate business insights using Python, SQL, and Power BI.

The project focuses on:

- Customer segmentation
- Loyalty analysis
- Revenue contribution
- Discount dependency
- High-value customer identification
- Strategic business recommendations

The workflow included:

1. Data Cleaning & Preprocessing
 2. Exploratory Data Analysis
 3. Feature Engineering
 4. SQL-Based Business Analysis
 5. Interactive Dashboard Development
 6. Strategic Recommendation & Playbook Creation
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2. Business Problem Statement

The primary objective of this project was to answer the following strategic business question:

“Is the business successfully building a loyal customer base, or is it reliant on continuous promotional activity? Furthermore, what strategic actions should be taken under either scenario?”

To answer this question, customer purchasing behavior, loyalty patterns, discount usage, subscription activity, repeat purchases, and revenue contribution were analyzed using Python, SQL, and Power BI.

3. Dataset Description

The dataset contains customer shopping behavior information including:

- Customer demographics
- Purchase amount
- Product category
- Payment method
- Subscription status
- Previous purchases
- Discount usage
- Location
- Loyalty indicators

Total records analyzed: 3900 customers

4. Data Cleaning & Preprocessing

The following preprocessing steps were performed:

- Removed duplicate records
- Standardized column names
- Handled missing values in review ratings
- Renamed inconsistent columns
- Converted categorical variables into usable formats
- Prepared data for feature engineering and visualization

These preprocessing steps ensured consistency, improved data quality, and enabled reliable business analysis.

5. Exploratory Data Analysis (EDA)

Several visualizations were created to understand customer behavior patterns.

Key Observations

Purchase Distribution

Purchase amounts were evenly distributed across customers without major outliers.

Category Performance

Clothing generated the highest revenue contribution, followed by Accessories.

Outerwear showed the weakest performance.

Gender Analysis

Male and female customers demonstrated similar spending behavior, indicating balanced purchasing trends across demographics.

Subscription Analysis

Subscribed customers demonstrated slightly stronger loyalty behavior and annual value.

Discount Analysis

Discount users did not significantly outperform non-discount users in purchase value or loyalty.

This suggests that heavy discounting may not strongly improve long-term customer retention.

6. Feature Engineering

New business-oriented features were created to improve customer intelligence and segmentation.

6.1 Loyalty Score

A weighted loyalty score was created using:

- Previous purchases
- Purchase amount
- Subscription status
- Discount dependency

The score was normalized and scaled between 0–100.

Why This Was Created

The loyalty score helps identify genuinely engaged customers instead of relying only on raw purchase value.

6.2 Customer Tier Segmentation

Customers were segmented into:

- Bronze
- Silver
- Gold
- Platinum

based on loyalty score ranges.

Why This Was Created

Tier segmentation allows businesses to:

- personalize marketing
 - prioritize retention
 - identify premium customers
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6.3 High Value Customer Flag

Customers with:

- strong loyalty
- high annual value
- repeat purchase behavior

were classified as high-value customers.

Why This Was Created

This helps businesses identify customers contributing the highest long-term business value.

6.4 Estimated Annual Value

Estimated yearly customer contribution was calculated using:

- purchase amount
- purchase frequency

Why This Was Created

Annual value estimation helps evaluate long-term profitability instead of single-order revenue.

7. SQL-Based Business Analysis

SQL queries were used to answer key business questions and generate actionable insights.

Key Questions Addressed

1. Who are the genuinely loyal customers vs. those who buy only during discounts?

Analysis Performed

- Loyalty score analysis
- Discount dependency analysis
- Promo code analysis
- Customer tier segmentation

Findings

- Platinum and Gold customers demonstrated genuine loyalty.
 - Non-discount customers generated slightly higher overall revenue.
 - Loyalty scores remained stable even without discounts.
 - Discounts did not significantly increase long-term customer value.
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2. What behavioral patterns predict high customer value over time?

Analysis Performed

- Repeat purchase analysis
- Loyalty score modeling
- Estimated annual value calculation
- High-value customer segmentation

Findings

High-value customers generally showed:

- higher repeat purchases
 - stronger loyalty scores
 - higher annual spending
 - lower discount dependency
 - subscription engagement
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3. Which geographies and demographics are commercially underleveraged?

Analysis Performed

- Location-based revenue analysis
- Category performance analysis
- Gender spending analysis

Findings

- Some states generated significantly higher revenue than others.
- Outerwear underperformed compared to Clothing and Accessories.
- Gender-based spending differences were minimal.

This helps identify underperforming regions and product segments for future growth opportunities.

4. How should the brand restructure its promotional strategy to protect margins without losing volume?

Analysis Performed

- Discount vs non-discount customer analysis
- Revenue contribution analysis
- Loyalty comparison across promotional groups

Findings

- Discount users did not significantly outperform non-discount users.
 - Heavy discounting may reduce margins without improving loyalty substantially.
 - Premium customers showed lower dependency on discounts.
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5. What does the brand's ideal customer profile look like?

Analysis Performed

- High-value customer analysis
- Tier-based segmentation
- Loyalty and annual value analysis

Findings

The ideal customer profile includes:

- Platinum-tier customers
 - high loyalty score
 - strong repeat purchase behavior
 - high annual spending
 - low discount dependency
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8. SQL Query Insights

Query 1 — Customer Distribution by Tier

Insight

Customer distribution across tiers was balanced, while Platinum customers generated the highest business value.

Query 2 — Average Spend & Loyalty by Tier

Insight

Platinum customers showed:

- highest average purchase value
- highest loyalty score
- strongest repeat purchase behavior

Bronze customers showed the weakest engagement metrics.

Query 3 — Revenue Contribution by Tier

Insight

Platinum customers contributed the highest total revenue, followed by Gold customers.

Query 4 — High Value Customer Analysis

Insight

Approximately 25% of customers qualified as high-value customers but contributed significantly stronger loyalty and annual value.

Query 5 — Discount Users vs Non-Discount Users

Insight

Discount users showed only marginal differences in purchase behavior compared to non-discount users.

Query 6 — Revenue Contribution by Discount Usage

Insight

Non-discount customers generated higher total revenue than discount users.

Query 7 — Customer Tier Distribution by Discount Usage

Insight

Premium customers showed lower dependency on discounts compared to lower-tier customers.

Query 8 — Promo Code Analysis

Insight

Promo code usage did not significantly improve loyalty or annual customer value.

Query 9 — Discount Dependency Risk Analysis

Insight

Loyalty remained relatively stable regardless of discount usage, especially among premium customers.

Query 10 — Top Revenue Locations

Insight

Several states demonstrated stronger customer value and revenue contribution than others.

Query 11 — Product Category Performance

Insight

Clothing and Accessories generated the highest revenue contribution.

Outerwear showed the weakest performance.

Query 12 — Subscription Analysis

Insight

Subscribed customers demonstrated:

- higher loyalty
 - higher annual value
 - stronger retention behavior
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Query 13 — Executive KPIs

Final KPIs

- Total Customers: 3900
 - Total Revenue: 233K
 - Average Order Value: 59.76
 - Average Loyalty Score: 46.66
 - High Value Customers: 25.08%
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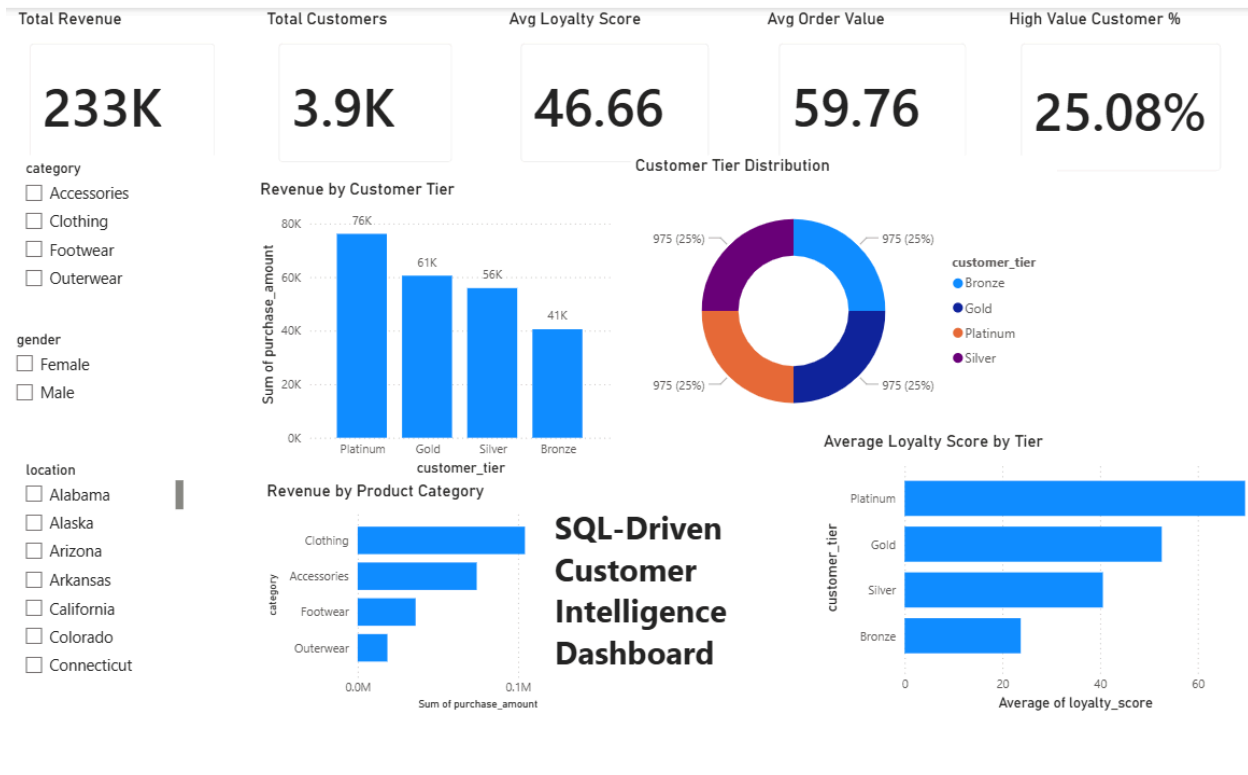
9. Power BI Dashboard

An interactive Power BI dashboard was developed to visualize:

- Revenue contribution
- Customer segmentation
- Loyalty analysis
- Category performance
- High-value customer insights

Dashboard Features

- KPI Cards
- Interactive Filters
- Revenue Comparison Charts
- Loyalty Score Analysis
- Customer Tier Analysis
- Product Category Analysis



10. Ideal Customer Profile (ICP)

The ideal customer identified through the analysis is:

- Platinum-tier customer
- Loyalty score above 60
- Frequent repeat purchases
- High annual spending
- Low dependency on discounts
- Strong subscription engagement

These customers contribute the highest long-term business value and retention potential.

11. Promo Sunset Playbook

Problem Statement

The analysis showed that discount-heavy strategies do not significantly improve customer loyalty or purchase behavior.

Heavy promotional dependency may reduce profitability without creating long-term retention.

Strategic Recommendations

1. Reduce Blanket Discount Campaigns

Gradually reduce universal discount campaigns to protect profit margins.

2. Introduce Personalized Promotions

Offer targeted rewards based on:

- loyalty score
 - customer tier
 - purchase history
 - repeat purchase behavior
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3. Prioritize Premium Customer Retention

Focus retention efforts on:

- Platinum customers
- Gold customers
- subscription users

4. Strengthen Subscription Programs

Enhance:

- membership rewards
- exclusive benefits
- loyalty incentives
- early-access programs

5. Expand High-Performing Categories

Increase marketing focus on:

- Clothing
- Accessories

due to stronger revenue contribution.

12. Strategic Business Conclusion

The analysis indicates that the business is NOT heavily dependent on promotional activity for customer retention.

Key Evidence

- Non-discount customers generated higher overall revenue.
- Loyalty scores remained stable even without discounts.
- High-value customers showed lower dependency on promotional offers.
- Subscription behavior and repeat purchases were stronger indicators of loyalty than discounts.

This suggests that long-term customer value is driven more by engagement and repeat behavior than aggressive promotional activity.

13. Executive Summary

This project successfully combined Python, SQL, and Power BI to build a complete customer intelligence system.

The analysis identified:

- high-value customer groups
- loyalty behavior patterns
- revenue-driving categories
- discount dependency risks
- customer retention opportunities

Major Findings

- Platinum customers generated the highest revenue and loyalty.
- Discounts had limited impact on long-term customer value.
- Clothing was the strongest-performing category.
- Only 25% of customers qualified as high-value customers.

Outcome

The project successfully delivered:

- a cleaned analytical dataset
 - engineered business features
 - SQL-driven business intelligence
 - an interactive Power BI dashboard
 - customer segmentation framework
 - strategic business recommendations
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14. Conclusion

This project demonstrates how customer analytics can help businesses:

- improve customer retention
- optimize promotions
- identify high-value users
- drive strategic decision-making

The integration of Python, SQL, and Power BI created a complete end-to-end analytics workflow capable of transforming raw customer data into actionable business intelligence.